```
# Attributes, columns, dimensions, variables   ------- (same name)
```

```
import pandas as pd
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings("ignore")
import seaborn as sns
```

```
df = pd.read_csv("ShopData.csv")
```

```
df.head()
```

	CustomerID	Genre	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

```
df.shape
```

```
(200, 5)
```

```
data = df.iloc[:, 3:5].values # used to extract as pairs
data
```

Show hidden output

```
data.ndim
```

```
2
```

```
sil_scores = []
scre = 0
kn = 2
for k in range(2,9):
  kmeans = KMeans(n_clusters=k, init="k-means++", random_state=5)
  label = kmeans.fit_predict(data)
  sil_score = silhouette_score(data, label)
  if scre < sil_score:
    scre = sil_score
    kn = k
  sil_scores.append(sil_score)
  print(f"Cluster: {k}; Silhouette Score: {sil_score}")
```

```
Cluster: 2; Silhouette Score: 0.3117892685561811
Cluster: 3; Silhouette Score: 0.38308008111229286
Cluster: 4; Silhouette Score: 0.4931963109249047
Cluster: 5; Silhouette Score: 0.5532176107575425
Cluster: 6; Silhouette Score: 0.5423120971494979
Cluster: 7; Silhouette Score: 0.5321706886465517
Cluster: 8; Silhouette Score: 0.49931779403714205
```

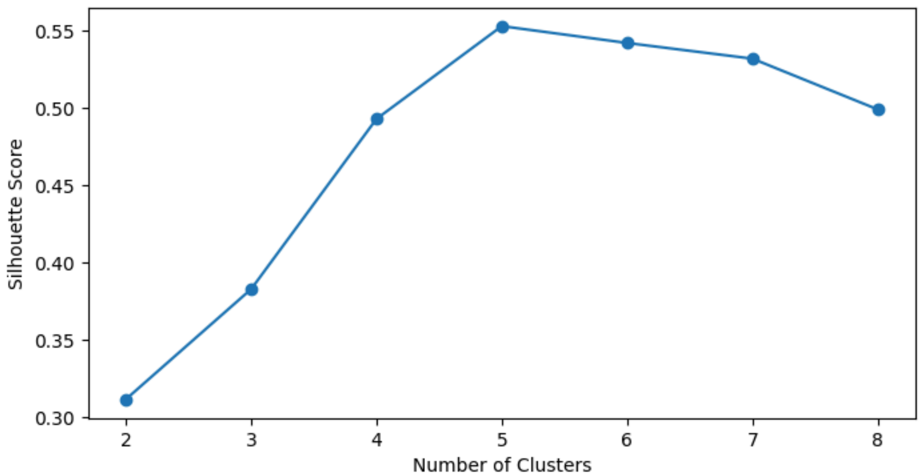
```
label
```

```
array([0, 4, 6, 4, 0, 4, 6, 4, 6, 4, 6, 4, 6, 4, 6, 4, 0, 4, 0, 4, 0, 4,
       6, 4, 6, 4, 0, 4, 0, 4, 6, 4, 6, 4, 6, 4, 6, 4, 0, 4, 0, 4, 0, 1,
       0, 4, 1, 0, 0, 0, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 5, 2, 5, 2, 5, 2, 5, 2, 5,
       2, 5, 2, 5, 2, 5, 2, 5, 2, 5, 1, 5, 2, 5, 2, 5, 2, 5, 2, 5, 2, 5,
       2, 5, 2, 5, 2, 5, 2, 5, 2, 5, 2, 5, 2, 5, 2, 5, 2, 5, 2, 5, 2, 5,
```

```
       2, 5, 2, 3, 2, 3, 2, 3, 2, 3, 2, 3, 2, 3, 2, 3, 7, 3, 7, 3, 7, 3,
       7, 3], dtype=int32)
```

```python
frame = pd.DataFrame({"Cluster" : range(2,9), "Silhouette Score" : sil_scores})
plt.figure(figsize=(8,4))
plt.plot(frame["Cluster"], frame["Silhouette Score"], marker="o")
plt.xlabel("Number of Clusters")
plt.ylabel("Silhouette Score")
```

```
Text(0, 0.5, 'Silhouette Score')
```

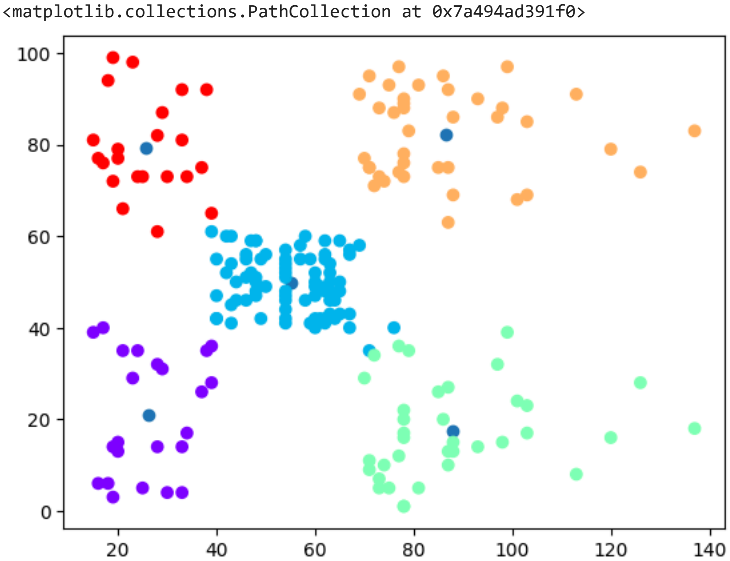



```python
kmns = KMeans(n_clusters=5, init = "k-means++", max_iter=300, random_state=5)
y_kmeans = kmns.fit_predict(data)
print(y_kmeans)
center = kmns.cluster_centers_
center
```

```
[0 4 0 4 0 4 0 4 0 4 0 4 0 4 0 4 0 4 0 4 0 4 0 4 0 4 0 4 0 4 0 4 0 4 0 4 0
 4 0 4 0 4 0 1 0 4 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
 1 1 1 1 1 1 1 1 1 1 1 1 3 2 3 1 3 2 3 2 3 2 3 2 3 2 3 2 3 2 3 1 3 2 3 2 3
 2 3 2 3 2 3 2 3 2 3 2 3 2 3 2 3 2 3 2 3 2 3 2 3 2 3 2 3 2 3 2 3 2 3 2 3 2
 3 2 3 2 3 2 3 2 3 2 3 2 3 2 3]
array([[26.30434783, 20.91304348],
       [55.0875    , 49.7125    ],
       [87.75      , 17.58333333],
       [86.53846154, 82.12820513],
       [25.72727273, 79.36363636]])
```

```python
plt.scatter(center[:,0], center[:,1], marker="o", )
plt.scatter(data[:,0], data[:,1], c=y_kmeans, cmap="rainbow")
```

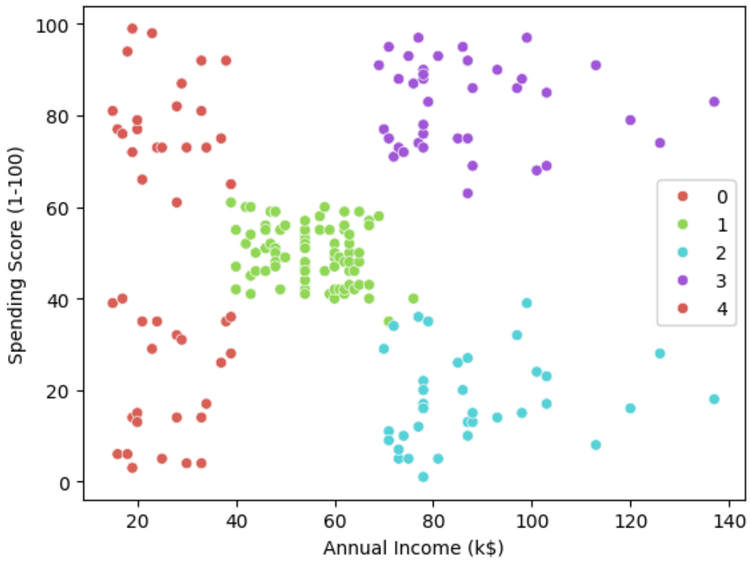
```
<matplotlib.collections.PathCollection at 0x7a494ad391f0>
```




```python
sns.scatterplot(data=df, x=df["Annual Income (k$)"], y=df["Spending Score (1-100)"], hue=y_kmeans
```

```
            , palette="hls")
```

<Axes: xlabel='Annual Income (k$)', ylabel='Spending Score (1-100)'>




```
clusterdf = pd.DataFrame({**df, "Cluster" : y_kmeans})
clusterdf.sort_values(by="Cluster")
```

	CustomerID	Genre	Age	Annual Income (k$)	Spending Score (1-100)	Cluster
0	1	Male	19	15	39	0
2	3	Female	20	16	6	0
6	7	Female	35	18	6	0
4	5	Female	31	17	40	0
12	13	Female	58	20	15	0
...	...	...	...	...	...	...
27	28	Male	35	28	61	4
19	20	Female	35	23	98	4
17	18	Male	20	21	66	4
21	22	Male	25	24	73	4
23	24	Male	31	25	73	4

200 rows × 6 columns

```
clusterdf
```

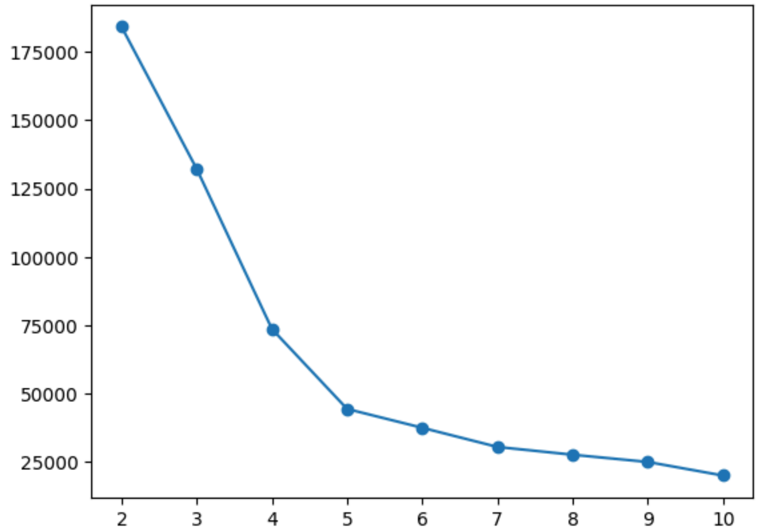
	CustomerID	Genre	Age	Annual Income (k$)	Spending Score (1-100)	Cluster
0	1	Male	19	15	39	0
1	2	Male	21	15	81	4
2	3	Female	20	16	6	0
3	4	Female	23	16	77	4
4	5	Female	31	17	40	0
...	...	...	...	...	...	...
195	196	Female	35	120	79	3
196	197	Female	45	126	28	2
197	198	Male	32	126	74	3
198	199	Male	32	137	18	2
199	200	Male	30	137	83	3

200 rows × 6 columns

```
SSE = []
clusMn = 2
clusMx = 11
for cluster in range(clusMn,clusMx):
    kmeans = KMeans(n_clusters = cluster, init='k-means++', random_state=5)
    kmeans.fit(data)
    SSE.append(kmeans.inertia_)
```

```
frame = pd.DataFrame({"Cluster" : range(clusMn,clusMx), "SSE" : SSE})
plt.plot(frame["Cluster"], frame["SSE"], marker="o")
```

```
[<matplotlib.lines.Line2D at 0x7a494a4b0ec0>]
```




```
Start coding or generate with AI.
```